

2026.2 Object Oriented Programming (class 04)

Project #1 [Due: Oct. 15 (Thursday) 11:59pm]

Submission

Case 1. If you are supposed to do project 1 as team project:

- Only the team leader should submit the project results (including source code files, execution demo video file in .mp4 format (length: around 1 or 2 minutes) and project report) through eclass system. A team member should give presentation. Please see and follow team project information and presentation schedule that will be available on our class webpage.)
- You should use English for project presentation and project report.

Case 2. If you are supposed to do project 1 as an individual project or your team size is 1,

- eClass submission – you should submit one zip file that contains two directories as follows
 - (i) directory "prob1" – this directory should contain a .pdf file that has only declaration of C++ classes with no function body implementation of IIKH system.
 - (ii) directory "prob2" – this directory should contain
 - source files (.cpp, .h, and solution files(.sln, and etc)) in visual studio format.
If you use g++, you should include Makefile that can be used for compilation.
Compilation and execution of your source files should be done successfully.
 - file1.txt you used
 - demo video file in .mp4 format (length: around 1 or 2 minutes) showing execution of your program including insertion, searching, and sorting (audio explanation in the video would be appreciated). You may create this demo video file by either using smart phone camera or screen capture.
Make sure to include the demo video file in your submission.
 - README.txt – very briefly explain how to compile and execute the source code.
(explain which OS and compiler (which version) you used, and how I can compile and execute your code (e.g. "click run button in Visual Studio", or execute "make")
- You should submit the above project results through eclass system before the deadline.

Problems

Prob#1. This problem is to perform Object Oriented Design of a system. Write C++ header files that contain class specification (with no function body implementation) for developing IIKH (Interactive Intelligent Kitchen Helper) that is described in Chapter 3 of T. Budd's book "Intro. to Object Oriented Programming". You need to just create classes that contain member variables and member functions. However, you do not need to implement function body of the member functions in prob1.

- In this problem, it is essential to choose a set of suitable classes and their member variables/functions that are necessary for IIKH.
- The names of classes, variables, and functions should reflect their meanings.
- You do not need to submit detailed implementation. However, you do have to insert appropriate comments in your class specifications.
- Above policies will be considered when grading your homework.
- The result of prob#1 should be one .pdf file that contains C++ classes.

Prob#2. The problem#2 is to fully develop a Student Information Management System. This program should support inserting/searching/displaying (sorted) student information. After the program is terminated, the inserted student information should be maintained. This requires you to use file processing. Do not use external library for the file processing. Use only C++ built-in libraries. You may use C++ STL library as well. The resulting C++ source code you submit must be compilable and executable.

Command Line Execution

The executable file takes one file as argument.

```
> a.exe file1.txt
```

a.exe is an executable file. file1.txt contains the student information saved during the program execution. You are free to organize the format of file1.txt. If file1.txt does not exist when execution, your program should create file1.txt. If file1.txt exists, you should use it.

(i) Main Menu

Following menu should be displayed when you execute the above command line.

```
1. Insertion
2. Search
3. Sorting Option
4. Exit
> _
```

(ii) If "1" (Insertion) is selected in the main menu, take input as follows and store the information into file1.txt.

Name ?
Student ID (10 digits)?
Birth Year (4 digits) ?
Department ?
Tel ?

"Name" has up to 15 (English) characters , "Student ID" should be exactly 10 digits where first 4 digits represent admission year. "Birth Year" should be exactly 4 digits. "Tel" has up to 12 digits. Department may contain a space character. After user's input is completed, the program should display the main menu and wait for user's input again. Name and Student ID should not be blank. If the same student id is provided, an error message "Error : Already inserted" should be printed.

(iii) If "2" (Search) is selected, display following search menu and take additional user input (keyboard).

- Search -
1. Search by name
2. Search by student ID (10 digits)
3. Search by admission year (4 digits)
4. Search by birth year (4 digits)
5. Search by department name
6. List All

> 5

Department name keyword? Computer Engineering

Name	StudentID	Dept	Birth Year	Tel
Lisa Simpson	2006303001	Computer Engineering	2000	01012345678
Gildong Hong	2004303077	Computer Engineering	1999	01187651234

After getting a number as user input in the above menu, your program should get additional string input for searching for student information stored in file1.txt

and display the search result. If you select "**6. List All**", informations for all students are displayed. The display order for student records should be based on the selection of "sorting option" (The default is "Sort by Name". This means that initially the display order should be sorted by name when no sorting option was specified). Go back to main menu after displaying the student records.

(iv) If "3" (Sorting Option) is selected , display following menu and take user input.

```
- Sorting Option
1. Sort by Name
2. Sort by Student ID
3. Sort by Birth Year
4. Sort by Department name
>
```

Go back to the main menu, after selecting the sorting option..

You may set the displaying order for printing out student information as a result of performing "Search" menu.

Notes

1. There is no deletion menu.
2. We don't consider program's efficiency. The correctness of your program will be mainly considered when grading your program.
3. you may consider using **qsort** function that is defined in `stdlib.h` or **sort** function in STL to implement "Sorting Option".